How the Temporal Fusion Transformer makes a forecast

Lim et al. (2021) · supervised baseline trained on each dataset · Darts TFTModel

1. A window of inputs

Each forecast reads a window of the last 120 days (36 months). A ? is a value not known yet.

past *future*

Target temperature

Known ahead day length

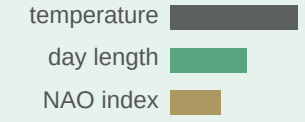
Past only NAO index

Prices go in as log differences, oil and yields as differences; all inputs are min-max scaled on the training rows.

Without known-ahead covariates a step counter is added as one.

2. Pick the useful inputs

At every step a small gated network gives each input a weight, so unhelpful inputs fade out.

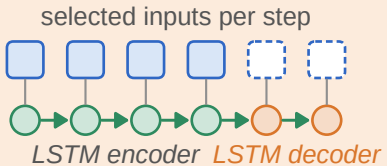


weights at one step (sum to 1)

Past steps weigh all inputs; future steps weigh only the known-ahead ones.

3. LSTM, then attention

An LSTM reads the past, a second one runs on over the horizon; then attention lets each future step look back over the past and the future steps before it.

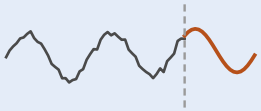


Attention: 2 heads that share one value projection, so the weights can be read

Gates and shortcuts around every part; hidden size 16 (tuned 8 to 32)

4. Train once, reuse

Trained once per dataset on the rows before the first test window (daily: the last 4,096), then every window is forecast from its own origin with no retraining.



point forecast

Optuna tries a few settings on validation windows inside the training rows; the first is the default. Trained here with MSE for a point forecast, not quantiles; 8 epochs.

Variable selection and attention make TFT one of the more readable deep forecasters.

1 LSTM layer · 2 attention heads · window 120 days or 36 months · direct forecast of 90 days or 24 months · tuned: hidden size, learning rate